Matthew Sooknah

Machine learning engineer with experience in computer vision, computational biology, and data engineering

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Experience

Calico Labs **South San Francisco, CA**
Staff Machine Learning Engineer *Feb 2025 – present*
Senior Machine Learning Engineer *August 2019 – Feb 2025*

- Built state-of-the-art machine learning models to extract high-dimensional spatiotemporal features from biomedical images and timeseries, and applied them to human cohort datasets like the UK Biobank to aid discovery of novel disease phenotypes and genetic targets.
- Led development of machine learning pipelines for object segmentation and quantification of disease signals across a wide range of imaging datasets spanning multiple organisms (cell culture, mice, humans) and organ systems (heart, lung, kidney, and vasculature). These pipelines have advanced key research and drug development programs, including one pipeline which is currently being validated and deployed in a clinical trial.
- Managed collaborations with biologists and imaging scientists to initiate new research projects, develop data collection and annotation protocols, and identify opportunities to use machine learning to save time / cost and increase predictive value of imaging data across Calico's research and drug development efforts.
- Worked with software engineers to productionize and automate data management, training and inference pipelines using Google Cloud Platform, Kubernetes and Airflow.
- Mentored a summer intern doing research on deep-learning techniques for CT image reconstruction, denoising, and domain-specific post-processing.
- Researched and applied the latest methods in self-supervised learning, generative models, temporal sequence modeling, and computer vision architectures like CNNs and ViTs.

Data Platform Engineer *August 2017 – July 2019*

- Built a data platform and web app for internal scientists to manage genomic samples, analyze and visualize results of experiments. Implemented server backend and data pipelines. Worked with UI designer to develop web-based GUI. Worked with scientists to collect requirements. These applications have been used for 3+ years to process thousands of experiments across basic research and drug development programs.
- Wrote pipelines to aid in assembling and publishing the genome sequence of an important cell-line (WI-38) from a variety of data sources. Implemented distributed pipeline for genome assembly, correction, and validation. Developed public-facing website and genome explorer to accompany the paper at wi38.research.calicolabs.com.
- Designed and co-taught a course on Python programming and data analysis for 30 internal wet-lab scientists, with the aim of empowering them to manage and analyze their own data without needing help from computational scientists. Students were able to apply what they learned to real data generated at Calico, increasing scientific productivity across the organization.

10X Genomics **Pleasanton, CA**
Scientist, Computational Biology *January 2016 – August 2017*

- Contributed to development of Cell Ranger and Long Ranger software packages, which are the industry standard for processing genomic data generated by the 10x microfluidics platform.
- Developed and refined algorithms for quality control, gene expression quantification, high-dimensional clustering, statistical analysis, and data visualization.

– Profiled and optimized performance of Python and Rust programs for I/O and memory-intensive applications.

The Broad Institute of MIT and Harvard **Cambridge, MA**
Software Engineer, Data Sciences & Data Engineering *May 2014 – December 2015*

– Developed and optimized analysis pipelines with Scala and Java for petabyte-scale genomics data from one of the largest DNA sequencing centers in the world.

– Contributed to development and support of multiple open source, industry standard tools for genome analysis (Picard and HTSJDK).

– Worked on backend database and data model for early prototype of cloud genome analysis platform (terra.bio).

– Developed methods for analyzing gene expression and pathway activity from customized RNA sequencing assays, to gain insight into the mechanisms of autoimmune disease.

Nabsys **Providence, RI**
Associate Scientist, Algorithms *June 2013 – May 2014*

– Developed tools in Java for signal processing and genome analysis to support development of the Nabsys microfluidics-based DNA mapping technology.

Education

Massachusetts Institute of Technology **Cambridge, MA**
S.B. Physics, GPA 4.9/5.0 *2009 – 2013*

Skills

– **Programming Languages:** Python (expert); Java, Scala, Rust, JavaScript, R, SQL (familiar)

– **Tools and Frameworks**: PyTorch, TensorFlow, Keras, sklearn, OpenCV, GCP, Docker, Airflow

Publications

– **Matt Sooknah**, Sivaramakrishnan Sankarapandian, Ramprakash Srinivasan, Johannes Riegler, Jun Xu. Video Representation Learning of Cardiac MRI for Genetic Discovery. Conference on Neural Information Processing Systems (NeurIPS), AIDrugX Workshop, December 2024.

– Sivaramakrishnan Sankarapandian, Ramprakash Srinivasan, **Matt Sooknah**, et al. Representation Learning based Target Discovery from UKBB MRI data. Conference on Neural Information Processing Systems (NeurIPS), AIDrugX Workshop, December 2024.

– Shivesh Chaudhary, Sivaramakrishnan Sankarapandian, **Matt Sooknah**, et al. Baikal: Unpaired Denoising of Fluorescence Microscopy Images using Diffusion Models. International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), October 2024.

– Ilya Soifer, Nicole Fong, Nelda Yi, Andrea Ireland, Irene Lam, **Matt Sooknah**, et al. Fully Phased Sequence of a Diploid Human Genome Determined de Novo from the DNA of a Single Individual. G3 (Genes, Genomes, Genetics), September 2020.

– Daniel O'Connell, Raivo Kolde, **Matt Sooknah**, et al. Simultaneous Pathway Activity Inference and Gene Expression Analysis Using RNA Sequencing. Cell Systems, May 2016.

Conference Talk Contributions

- Florian Schmid, Georgios Koukos, **Matt Sooknah**, et al. "Kidney T2 can be used as a biomarker for disease progression and functional decline in mouse models of polycystic kidney disease." International Society for Magnetic Resonance in Medicine (ISMRM), June 2023.
- Florian Schmid, Georgios Koukos, **Matt Sooknah**, et al. "Tracking disease progression in mouse models of polycystic kidney disease with high resolution MRI and automated postprocessing." International Society for Magnetic Resonance in Medicine (ISMRM), May 2022.
- Florian Schmid, Georgios Koukos, Yi Liu, **Matt Sooknah**, et al. "High-resolution kidney MRI in mice for longitudinal tracking of kidney volume and cyst burden." International Society for Magnetic Resonance in Medicine (ISMRM), May 2021.
- Grace Zheng, Jessica Terry, Paul Ryvkin, **Matt Sooknah**, et al. "Single Cell RNA profiling of a Million Neurons by a Massively Parallel and Scalable Droplet Platform." Advances in Genome Biology and Technology (AGBT), February 2017.
- Haynes Heaton, Patrick Marks, **Matt Sooknah**, et al. "Alignment and Variant Calling in Segmental Duplications with Linked-Read Data." Genome Informatics, September 2016.